

Beyond Access: Does Usage Intensity Predict Exposure to Digital Financial Fraud?

Part 3 – Digital Inclusion in Italy (IACOFI 2023)

Abstract

Having established in Part 1 that a substantial share of the population is not online at all, this part narrows the focus to the population that *is* digitally active and asks a different question: among digitally engaged respondents, does the intensity of digital financial service use predict exposure to scams, phishing attempts, or unauthorised transactions? Using a purpose-built usage-intensity index and a companion digital-skill index, logistic regression models show that more intensive users report substantially more incidents – medium-intensity users are almost 3.8 times more likely to report one than non-users – and this association survives adjustment for age, education, income and geography. Critically, when digital skill is added to the model, both usage intensity *and* skill remain independently and positively associated with incident risk. This rules out a simple “low competence causes victimisation” account and instead points to *exposure*: using digital financial services more often mechanically creates more occasions for something to go wrong. Raw incident counts should therefore not be read as a proxy for skill deficits.

1 Research Question

Part 1 asked *who is excluded* from the digital environment altogether. Part 3 asks a question that only makes sense for those who have already cleared that hurdle: **are more intensive users of digital financial services also more likely to report scams, phishing, or unauthorised transactions?** A naive reading of a positive correlation between usage and incidents would conclude that heavy users are simply worse at protecting themselves – that incidents are a symptom of low digital competence. This part treats that inference as a hypothesis to be tested rather than a starting assumption: because greater activity mechanically creates more opportunities for a scam or unauthorised transaction to occur, a positive association between usage intensity and incident rates is exactly what a pure *exposure* mechanism would also predict, independently of any skill deficit. Distinguishing between these two explanations – exposure versus competence – is the analytical task of this part.

2 Feature Construction

Three variables specific to this part were built from the questionnaire, in addition to the age, education, income and urbanisation groupings already introduced in Part 1, which are reused here unchanged as controls.

Outcome: `digital_incident`. Four questionnaire items record distinct adverse events: a suspicious transaction, a phishing attempt, unauthorised account access, and any other fraud attempt. Rather than modelling each separately, `digital_incident` is built as the row-wise maximum across the four, so it is 1 if the respondent reports *at least one* of them and 0 otherwise. This “any incident” framing keeps the outcome simple to interpret and statistically well powered; Section 3 still reports the four components separately (Figure 4) so that the pooled result is not hiding a single idiosyncratic item.

Key predictor: `digital_finance_intensity`. Seven items ask how often the respondent uses distinct digital financial behaviours (e.g., mobile payments, online transfers, digital wallets). Their row-wise mean produces a continuous index running from 1 (never) to 4 (frequent use across the board), which is the natural way to combine several correlated frequency items into a single exposure scale without arbitrarily privileging any one channel. For presentation purposes the same index is also cut into four ordered usage groups – non-user, low use, medium use, and intense use – at the 1.5, 2.5 and 3.5 breakpoints, which lets the descriptive figures speak in plain group labels while the regression models use the finer continuous version.

Competence check: `digital_skill`. Six further items measure self-assessed digital literacy (e.g., comfort using apps, recognising unfamiliar interfaces) and are averaged the same way into a second continuous index. This index exists purely to arbitrate between the two competing explanations set out in Section 1: if incidents were primarily a competence problem, `digital_skill` should enter any model *negatively* once usage intensity is held constant, and the usage-intensity coefficient itself should shrink once skill is controlled for. Both of these predictions are tested directly in Section 4.

Modelling strategy. Three logistic regressions were estimated with `digital_incident` as the dependent variable: (1) a bivariate model with `digital_finance_intensity` alone; (2) a model adding the age, gender, education, income, employment and urbanisation controls; and (3) a model that further adds `digital_skill`. This sequence isolates, step by step, whether the usage–incident link is confounded by who uses digital finance, and, decisively, whether it survives – or is explained away by – a direct measure of digital competence.

3 Descriptive Evidence

Across the sample, 720 of 4,742 respondents with valid data (15.2%) reported at least one digital financial incident.

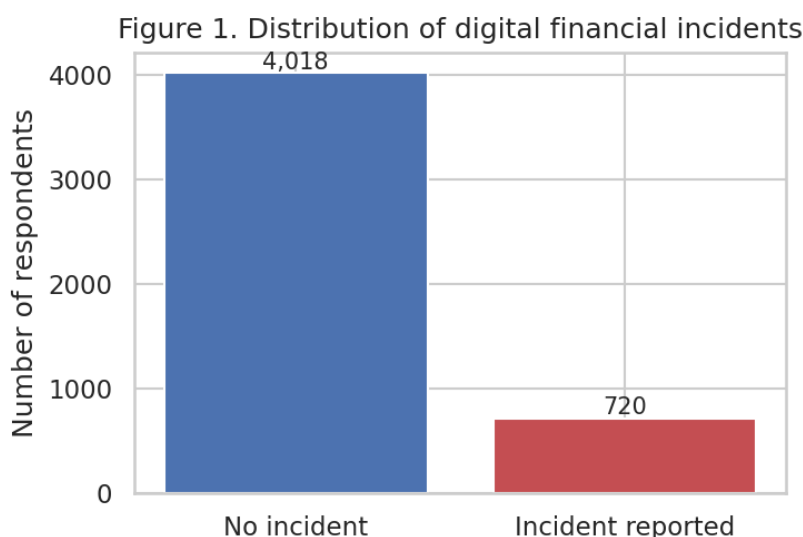


Figure 1: Distribution of digital financial incidents across the sample.

Incident prevalence rises sharply and close to monotonically with usage intensity, before dipping slightly in the smallest, most intense-use segment (Figure 2): 7.9% of non-users report an incident,

versus 12.4% of low-intensity users, 29.7% of medium-intensity users, and 20.6% of the intense-use group ($n = 102$, the smallest of the four cells).

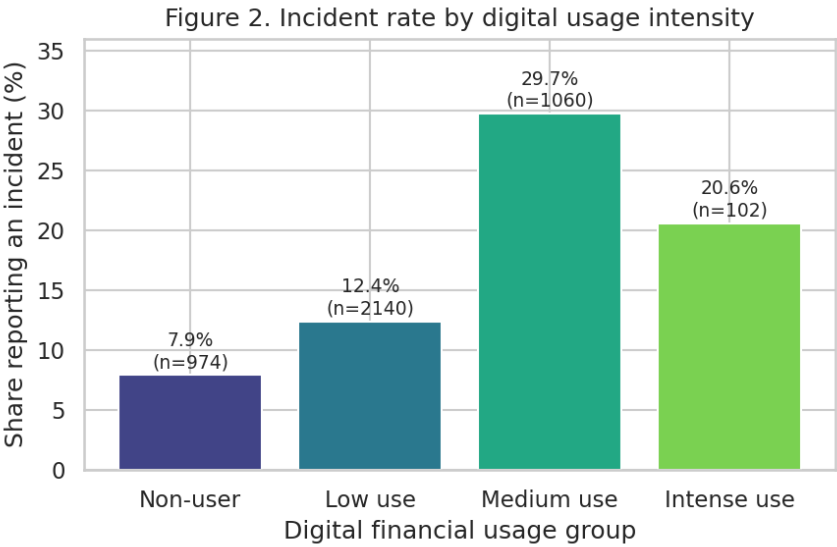


Figure 2: Share of respondents reporting a digital financial incident, by usage-intensity group.

Expressed as risk relative to non-users, low-intensity users face 1.6 times the risk, medium-intensity users face 3.8 times the risk, and intense users face 2.6 times the risk (Figure 3). The dip at the very top of the usage distribution is more plausibly a small-sample artefact – or a sign that the most active users are also the most experienced – than evidence that heavy usage becomes protective; this is examined further in Section 5.

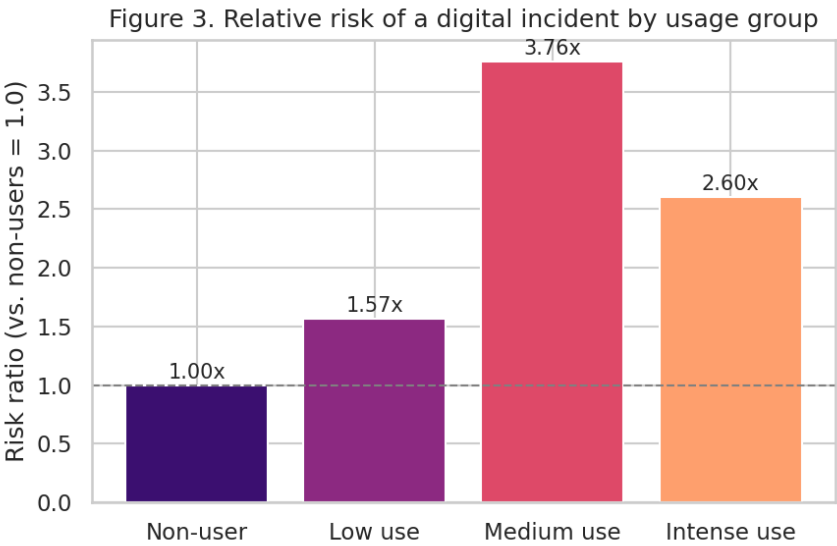


Figure 3: Incident risk by usage group, expressed relative to non-users (=1.0).

Breaking the outcome down into its four components shows the usage gradient is not driven by a single kind of event (Figure 4): suspicious transactions, phishing attempts, unauthorised access and other fraud attempts are all more prevalent among medium- and intense-use respondents than among non-users and low-intensity users. Unauthorised access – a largely passive event from the victim’s point of view – follows the same gradient as the more behaviourally-mediated categories, which already hints that the pattern is about exposure rather than user error.

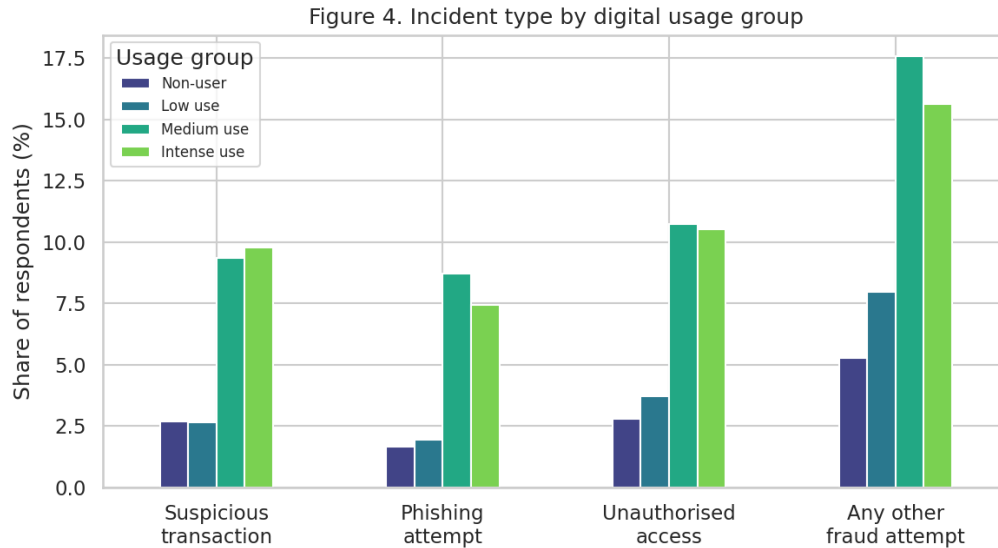


Figure 4: Prevalence of each incident type, by digital usage group.

4 Regression Results

The bivariate model confirms that the usage–incident association is statistically robust: a one-unit increase in `digital_finance_intensity` is associated with an odds ratio of roughly 2.35 for reporting an incident ($\beta = 0.853$, $SE = 0.063$, $z = 13.53$, $p < 0.001$; pseudo- $R^2 = 0.051$, $n = 4,276$; Figure 5).

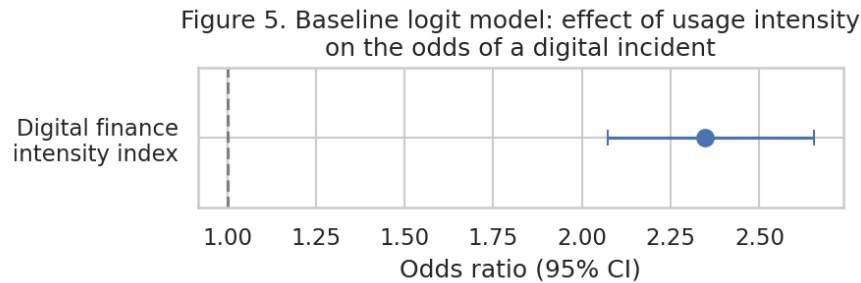


Figure 5: Odds ratio for digital financial incidents associated with a one-unit increase in usage intensity (bivariate model).

Adding the demographic controls leaves the intensity effect essentially unchanged ($\beta = 0.856$, $SE = 0.070$, $z = 12.26$, $p < 0.001$; pseudo- $R^2 = 0.093$, $n = 4,259$). Two secondary patterns emerge alongside it: incident risk rises with age, and respondents in the middle income bracket show significantly higher odds than the highest earners, while residents of mid-sized urban centres show somewhat lower odds than residents of the largest cities. Neither of these secondary patterns changes the substantive reading of the intensity effect.

The decisive test is the third model, which adds `digital_skill` to the intensity model. If low competence, rather than exposure, were driving the pattern, `digital_skill` should enter with a *negative* coefficient, and the intensity effect should shrink once skill is controlled for. Neither prediction holds (Figure 6). The intensity effect strengthens slightly ($\beta = 1.169$, $SE = 0.078$, $z = 15.00$, $p < 0.001$, odds ratio ≈ 3.22), and `digital_skill` is itself positive and highly significant ($\beta = 0.730$, $SE = 0.082$, $z = 8.89$, $p < 0.001$, odds ratio ≈ 2.07). Pseudo- R^2 rises to 0.112 ($n = 4,268$), the best fit of the three models.

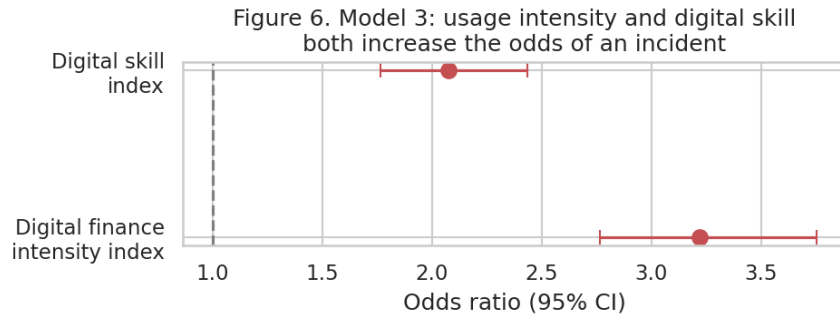


Figure 6: Odds ratios for usage intensity and digital skill in the fully adjusted model. Both point estimates lie to the right of 1, indicating higher – not lower – incident odds for more skilled and more active users.

Model	Intensity β (OR)	Skill β (OR)	Pseudo- R^2	n
1. Bivariate	0.853 (2.35)	–	0.051	4,276
2. + demographics	0.856 (2.35)	–	0.093	4,259
3. + digital skill	1.169 (3.22)	0.730 (2.07)	0.112	4,268

Table 1: Summary of the three logistic models of `digital_incident`. Odds ratios (OR) in parentheses. Full demographic controls (age, gender, education, income, employment, urbanisation) are included in Models 2 and 3.

5 Discussion: Exposure, Not Competence

The results answer the research question directly: yes, more intensive users of digital financial services are considerably more likely to report scams, phishing attempts, or unauthorised transactions, and the association is not an artefact of who uses digital finance – age, income, education and geography do not explain it away.

However, this pattern should not be read as evidence that heavy users are less competent or more careless. Three pieces of evidence argue against a competence-deficit explanation. First, `digital_skill` is positively – not negatively – associated with incident risk in Model 3, the opposite of what a skill-deficit story would predict. Second, the intensity effect does not weaken once skill is controlled for; if low competence explained both high usage and high incident risk, adjusting for skill should have attenuated the intensity coefficient, and it did not – it strengthened. Third, the gradient in Figure 4 holds across every incident type, including unauthorised access, which is less plausibly explained by user error than by simply having more accounts, cards or channels that could be targeted.

Taken together, the evidence is best explained by *exposure*: a respondent who uses several digital financial channels every week has proportionally more transactions, logins and financial touchpoints than a non-user, and therefore more occasions on which a criminal attempt can reach them, or an anomaly be flagged. More skilled users may also be more likely to correctly recognise and label an attempted scam as such, rather than dismissing it – which could itself inflate reported rates among high-skill respondents independently of any true difference in underlying victimisation. This measurement dynamic is a further reason to treat the skill coefficient as evidence of *exposure and awareness*, not of vulnerability.

The slight dip in incident rates among the smallest, most intense-use segment (Figures 2–3) is also more consistent with an exposure/experience account than a competence-deficit one: the most active users may combine the highest exposure with the greatest platform familiarity, partially

offsetting risk once usage moves beyond "medium" levels – though with only 102 respondents in that cell this remains a descriptive observation rather than a robust finding.

6 Synthesis

Three conclusions follow directly from this part of the analysis:

1. Digital financial usage intensity is a strong, robust predictor of reporting a digital financial incident, surviving adjustment for demographic composition.
2. Digital skill does *not* act as a protective factor in these data – if anything, it is positively associated with incident reporting, most plausibly because active, skilled users are also the most exposed and the most able to recognise an incident when it occurs.
3. Consumer-protection resources should therefore be targeted by *exposure* (frequency and breadth of digital financial activity) rather than by an assumption that vulnerable users are unsophisticated or infrequent users; the population most exposed to digital financial fraud is, in this sample, the population that is already digitally active.

Read alongside Part 1, the two parts describe a two-stage funnel: a first divide determines who is online at all – concentrated among older and less-educated respondents – and, conditional on being online and financially active, a second gradient determines who is exposed to fraud, driven not by a lack of skill but by the sheer intensity of digital engagement itself.

Limitations

- The data are cross-sectional, so the direction of causality between usage intensity and incidents cannot be established; experiencing an incident may itself alter subsequent usage behaviour.
- The outcome captures any *reported* incident, not confirmed financial loss, and may partly reflect respondents' ability to recognise an event as suspicious – itself correlated with digital skill.
- The intense-use segment is small ($n = 102$), which limits the precision of estimates specific to that group.
- Pseudo- R^2 remains modest (0.05–0.11) across all models, so usage intensity and skill, while robust predictors, leave most of the variation in incident reporting unexplained.